
CY 101: Chemistry

Course Code	CY 101
Course Name	Chemistry
L-T-P-C	3-1-2-5
Prerequisites	Nil
Distribution	Core
One-time approval	5th Senate (Oct 2013)

Course Contents

- **Quantum Mechanics**
Postulates, Schrödinger wave equation, Interpretation of wave function, particle in a box, implication of the concepts, H atom, radial and angular wave functions, and shapes of orbital.
- **Thermodynamics**
Statistical concept of entropy, free energy, and chemical equilibria, chemical potential.
- **Kinetics and Catalysis**
Theories of chemical reactions, homogeneous and heterogeneous catalysis.
- **Electrochemical Cells**
Electrochemical corrosion and fuel cells.
- **Stereoisomerism**
Overview of concepts, configuration, Fischer and Newman projections. Optical isomerism of simple cyclic systems and of compounds without asymmetric carbon atom (allenes, spirocompounds, etc.); chirality involving atoms other than carbon.
- **Reaction Mechanism and Stereochemistry in Organic Synthesis**
Synthesis: (a) Addition of KMnO_4 , OsO_4 , and peracids to cis- and trans alkenes (b) Diels-Alder reaction: (4+2) cyclo addition MO treatment (c) Aromatic Nucleophilic substitution mechanism ($\text{S}_{\text{N}}\text{Ar}$, $\text{S}_{\text{N}}1$, Arynes) reactivity and reactions;
- **Novel Polymers** Stereo chemical control of synthesis, Ziegler-Natta catalyst, Polyurethanes, conducting polymers
- **Spectroscopic Techniques**
Introductory ideas of molecular spectroscopy and applications of UV visible and IR to simple compounds/coordination complexes.
- **Coordination Compounds**
Crystal field theory of octahedral and tetrahedral complexes, colour and magnetic properties, Jahn-Teller distortion with specific reference to d9 case.
- **Organometallics**
(i) Metal carbonyls: synthesis, structure and bonding (ii) Metal alkene complexes: bonding in metal alkene complexes, role of metal alkene complexes in hydrogenation and hydroformylation.
- **Metal ions in Biological Systems**
Role of trace metals in biological systems with special reference to transition metals (Cu, Fe, Zn), toxic effects of Cd and Hg.

Laboratory Modules

1. Determination of iron in iron ore using potassium dichromate (Internal indicator method);
2. Determination of sodium carbonate in baking/washing soda;
3. Determination of hardness of water by EDTA- complexometry titrations;
4. Heat of neutralization of a strong base by a strong acid;
5. Equivalent weight of an acid;
6. Viscosity of mixtures of liquids;
7. Surface excess of 1-butanol in aqueous solution;
8. Order of reaction;
9. Percentage of ammonia in an ammonium salt;
10. Identification of functional groups in organic compounds;
11. Blue Printing
12. pH metry / potentiometry titrations; a) Strong acid – strong base; b) Strong acid – weak base; c) Weak acid – strong base; d) Redox titration: Fe^{2+} or Mn^{2+} ;
13. Spectro photometry: Determination of Fe(III) by colorimetry;
14. Determination of water of crystallization by microwave irradiation;
15. Preparation of acetanilide or aspirin and determination of melting point, and matching with known sample;
16. Determination of chloride of As or Cr in Water.

References

1. Lee J.D., **Concise Inorganic Chemistry**, 5th Edition, Chapman & Hall.
2. Malik T. and Madan, **Selected Topics in Inorganic Chemistry**, 5th Edition, S.Chand & Company.
3. Peter S., **A guide book to Mechanism in Organic Chemistry**, 6th Edition, Orient Longman.
4. Morrison R.T. and Boyd R.N., **Organic Chemistry**, 6th Edition, Printice Hall of India.
5. Mahan B.H., **University Chemistry**, 3rd Edition, Narosa Publishing House.
6. Atkins P.W., **Physical Chemistry**, 5th Edition, ELBS, Oxford University Press.